AI / ML Engineer Interview Guide

Original practice prompts and concise answer frameworks

A production-oriented guide across model development, evaluation, deployment, monitoring, ML systems, and LLM applications. Designed for retrieval by a resume-based interview preparation app.

How to use this guide

Treat each item as a practice prompt, not a script. Answer aloud, anchor claims in your own resume, and adapt the depth to the role and seniority. The sample answers are concise frameworks; add your own evidence and tradeoffs.

1. Walk through an ML system from request to production.

Answer framework Start with the user problem, prediction target, latency and reliability needs, and a simple baseline. Define data contracts and point-in-time features, train and validate with deployment-like splits, then package preprocessing with the model. Serve through batch or online inference, log inputs and outcomes safely, monitor quality and drift, and define rollback, retraining, and ownership.

2. How do you detect and respond to data drift?

Answer framework Track input distributions, missingness, schema, prediction distributions, and delayed outcome metrics by important segment. Distinguish covariate shift from changes in the target relationship where possible. Alerts should be tied to validated thresholds and impact. Investigate upstream changes, evaluate on fresh labeled data, and retrain only after confirming the cause and regression tests.

3. What is the difference between batch and online inference?

Answer framework Batch inference precomputes predictions for many records and suits scheduled decisions with relaxed freshness needs. Online inference serves each request with low latency and needs reliable feature retrieval, capacity planning, and fallback behavior. Some systems combine both. Choose based on freshness, cost, traffic, and failure tolerance.

4. How would you handle training-serving skew?

Answer framework Use shared transformation code or a feature store with consistent definitions, point-in-time joins, and versioned schemas. Log a sample of serving features and compare them with offline values for the same entities and timestamps. Add parity tests and monitor missing or out-of-range features. Make transformations deterministic and avoid future information.

5. How would you deploy a model safely?

Answer framework Package a reproducible artifact with preprocessing and dependencies, validate it in staging, and compare against the current system using shadow traffic or a small canary. Define latency, error, quality, and business guardrails before rollout. Keep a versioned rollback path and test it. Expand traffic only when observed results remain within limits.

6. What is concept drift?

Answer framework Concept drift is a change in the relationship between inputs and the target, so a previously learned mapping becomes less useful. It differs from a change in input distributions alone. Detect it with outcome-based metrics when labels arrive, plus proxy signals meanwhile. Investigate regime changes and retrain using representative recent data after validation.

7. How do you choose a model serving architecture?

Answer framework Estimate request volume, latency budget, model size, update cadence, privacy needs, and availability targets. A managed endpoint can simplify operations; a containerized service offers control; batch jobs reduce per-request overhead. Benchmark end-to-end latency and cost, include feature lookup and serialization, and design timeouts and fallbacks.

8. What should you monitor after deployment?

Answer framework Monitor service health (latency, throughput, errors, resource use), data quality and feature distributions, prediction behavior, and delayed model outcomes. Slice by important populations and business segments. Use actionable thresholds, ownership, and runbooks. Alert fatigue is reduced when each alert corresponds to a decision or remediation.

9. How would you design a recommendation system?

Answer framework Clarify the objective, context, inventory, and constraints such as freshness, diversity, and safety. A common design retrieves candidates cheaply, ranks them with richer features, then applies business and policy constraints. Evaluate offline with temporal splits and ranking metrics, then run online experiments with guardrails. Plan feedback loops and cold-start handling.

10. How do you serve a large language model feature?

Answer framework Define the task, quality bar, latency and cost budgets, and allowed data. Establish a baseline using prompting or retrieval before fine-tuning. For retrieval, chunk and index trusted content, retrieve relevant passages, and ground answers with citations. Evaluate task-specific quality, unsafe or unsupported responses, latency, and cost; use caching, fallback, and human review where appropriate.

11. What is a feature store useful for?

Answer framework It can centralize feature definitions, reuse, lineage, and online/offline access while helping keep training and serving values consistent. It is useful when multiple models or teams share features and freshness requirements. It adds operational complexity, so begin with clear ownership, point-in-time correctness, and a measurable need.

12. How would you reduce inference cost?

Answer framework Measure cost by request and identify whether compute, model size, network, or feature retrieval dominates. Consider batching, caching, quantization, distillation, smaller models, or routing simple cases to cheaper models. Benchmark quality and tail latency after each change, and keep guardrails for cases where the cheaper path fails.

Research references

These references informed the topic selection and preparation emphasis. The prompts and answer frameworks in this PDF are newly written and paraphrased for study use; they are not reproductions of the linked articles.

- **GeeksforGeeks - AI ML DS Interview Series**
https://www.geeksforgeeks.org/interview-experiences/ai-ml-ds-interview/

- **Medium / GrabNGoInfo - Top 60 Machine Learning Interview Questions**
https://medium.com/grabngoinfo/top-60-machine-learning-interview-questions-5f8baafc76ed

- **Microsoft Careers - Technical interviewing**
https://careers.microsoft.com/v2/global/en/hiring-tips/technical-interviewing

Knowledge-base note: this material is educational interview practice, not a guarantee of interview coverage. Tailor preparation to the role description, level, and candidate resume.